You should Plugin your Telemed Ultrasound

Pertti Palo[1], Co-author Full Name[2]

[1]University of Alberta, Edmonton, Alberta, Canada
[2]Affiliation

pertti.palo@taurlin.org, Author 2 email address

Introduction. Ultrasound has been used for speech studies for decades (Shawker & Sonies, 1985; Stone, 2005) In recent years Telemed micro ultrasound devices have become something of a norm **[cite 3-4 studies]**. While the manual says to use mains power (connect the device's power source to a wall socket) it does not say why, and the device will work with just power from the USB-connection to a computer. We discovered by accident why this recommendation exists: When running on USB power the ultrasound device will raise imaging power by steps during each recording. This is done to avoid setting off a short circuit panic mode in the USB port. Unfortunately, the stepping of intensity has consequences for data quality. This presentation provides a description of the consequences, a description of how to detect if old data has been compromised and a recommendation for future data recording.

Method.

- PD (Palo, 2019)
- PATKIT (Palo, Moisik, & Faytak, 2026)

Results.

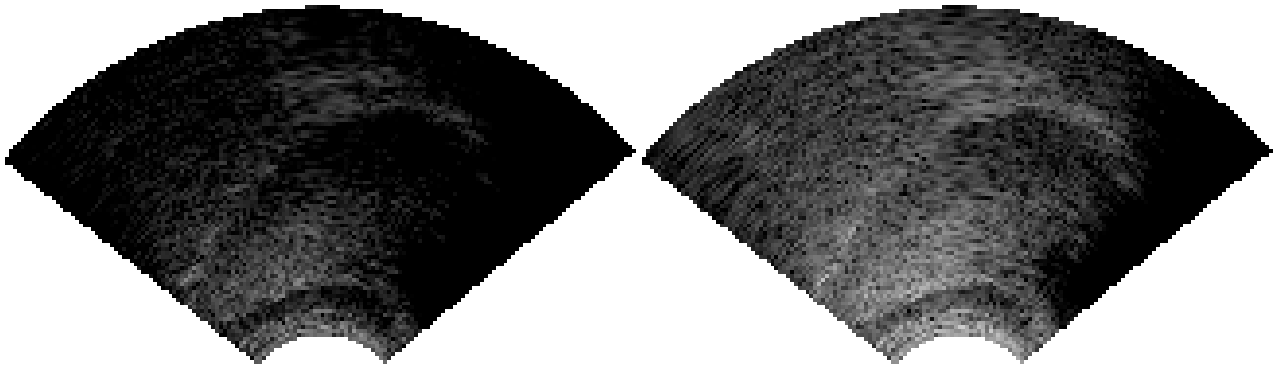


Figure 1: foo.

- example of bad data in PD

Conclusion. For new data the recommendation is simple: Plug the ultrasound machine into mains power when ever possible and if you need to run it with only USB power, make a note of that in your recording metadata / laboratory journal.

For old data, if you do not know that the device was plugged in to mains electricity, there are two ways of checking if there is a problem with data quality. First, play a recording and look at the data.

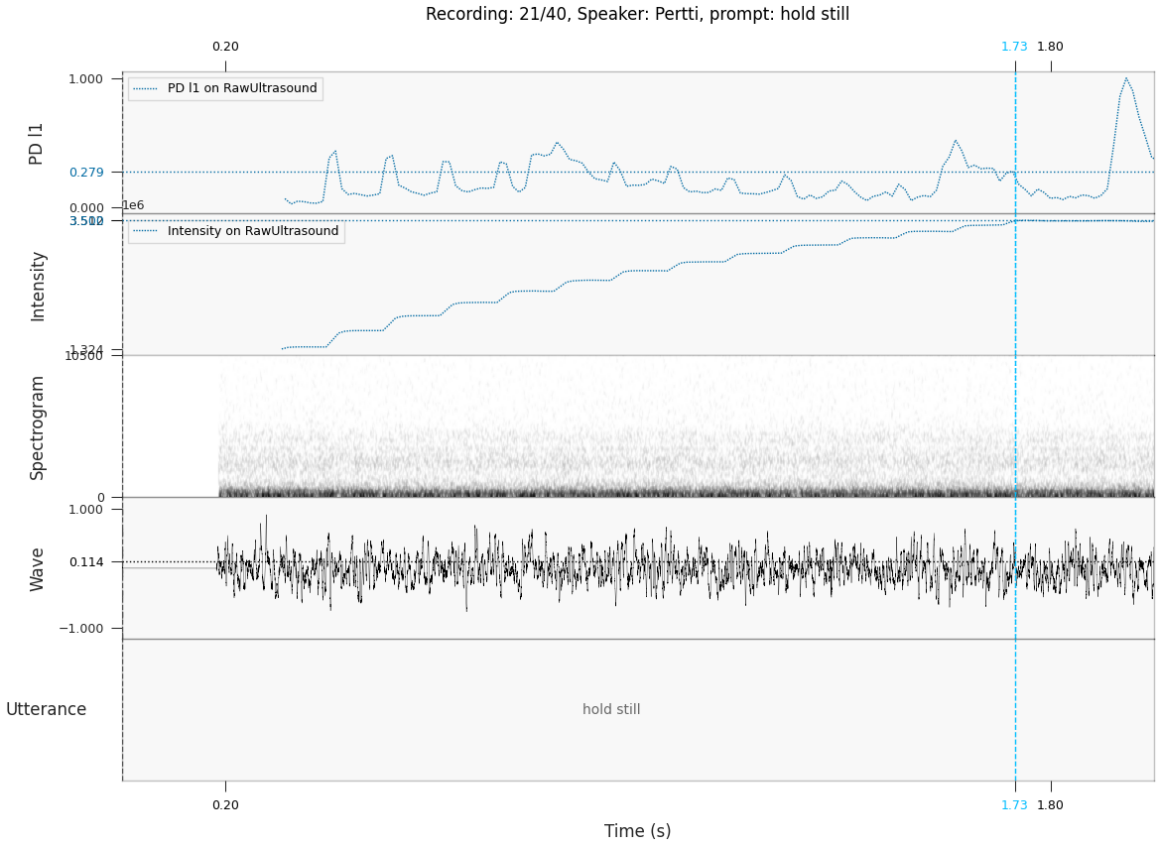


Figure 2: foo.

If the ultrasound image seems to get brighter they may be a problem. Second, examine the PD curves in AAA or PATKIT. You may need to look at more than one recording to see if intensity is going up step-wise like in Figure 2. In case the intensity is ramping up, any intensity based measures might be unreliable.

References.

Palo, P. (2019). *Measuring pre-speech articulation* (Doctoral dissertation, Queen Margaret University, Edinburgh). Retrieved 2026-08-28, from `https://eresearch.qmu.ac.uk/items/dc26c3ed-5467-4f8c-a1c7-f82b53db6084`

Palo, P., Moisik, S. R., & Faytak, M. (2026). *PATKIT: Phonetic Analysis ToolKIT. [Python software package].* Available in a public software repository, accessed 20 March 2026.

Shawker, T. H., & Sonies, B. C. (1985). Ultrasound biofeedback for speech training. Instrumentation and preliminary results. *Investigative Radiology*, *20*(1), 90–93. 10.1097/00004424-198501000-00022

Stone, M. (2005). A Guide to Analyzing Tongue Motion from Ultrasound Images. *Clinical Linguistics and Phonetics*, *19*(6–7), 455–502.